



MODULE ONE PROBLEM SET

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Tyler Ellis



Directions: Type your solutions into this document and be sure to show all steps for arriving at your solution. Just giving a final number may not receive full credit.

PROBLEM 1

- (a) Every patient was given the medication or the placebo or both.

$$\forall x (D(x) \vee P(x)).$$

$$\neg \forall x (D(x) \vee P(x)) \equiv \exists x \neg (D(x) \vee P(x)) \equiv \exists x (\neg D(x) \wedge \neg P(x)).$$

English: There is at least one patient who was given neither the medication nor the placebo.

- (b) Every patient who took the placebo had migraines. (Hint: you will need to apply the conditional identity, $p \rightarrow q \equiv \neg p \vee q$.)

$$\forall x (P(x) \rightarrow M(x)) \equiv \forall x (\neg P(x) \vee M(x)).$$

$$\neg \forall x (\neg P(x) \vee M(x)) \equiv \exists x \neg (\neg P(x) \vee M(x)) \equiv \exists x (P(x) \wedge \neg M(x)).$$

English: There is at least one patient who took the placebo and did not have migraines.

- (c) There is a patient who had migraines and was given the placebo.

$$\exists x (M(x) \wedge P(x)).$$

$$\neg \exists x (M(x) \wedge P(x)) \equiv \forall x \neg (M(x) \wedge P(x)) \equiv \forall x (\neg M(x) \vee \neg P(x)).$$

English: Every patient either did not have migraines or was not given the placebo.



PROBLEM 2

Use De Morgan's law for quantified statements and the laws of propositional logic to show the following equivalences:

(a)

$$\neg \forall x (P(x) \wedge \neg Q(x)) \equiv \exists x \neg (P(x) \wedge \neg Q(x)) \equiv \exists x (\neg P(x) \vee Q(x)).$$

(b)

$$\neg \forall x (\neg P(x) \rightarrow Q(x)) \equiv \neg \forall x (P(x) \vee Q(x)) \equiv \exists x \neg (P(x) \vee Q(x)) \equiv \exists x (\neg P(x) \wedge \neg Q(x)).$$

(c)

$$\neg \exists x (\neg P(x) \vee (Q(x) \wedge \neg R(x))) \equiv \forall x \neg (\neg P(x) \vee (Q(x) \wedge \neg R(x))) \equiv \forall x (P(x) \wedge (\neg Q(x) \vee R(x))).$$



PROBLEM 3

The domain of discourse for this problem is a group of three people who are working on a project. To make notation easier, the people are numbered 1, 2, 3. The predicate $M(x, y)$ indicates whether x has sent an email to y . The table below shows the value of the predicate.

M	1	2	3
1	T	T	T
2	T	F	T
3	T	T	F

Determine if each quantified statement is true or false. Justify your answer.

(a)

$$\forall x \forall y (x \neq y \rightarrow M(x, y)).$$

Answer: True. **Justification:** For every pair $x \neq y$ among $\{1, 2, 3\}$, $M(x, y) = T$ in the table, so each implication holds.

(b)

$$\forall x \exists y \neg M(x, y).$$

Answer: False. **Justification:** For $x = 1$, $M(1, 1) = T$, $M(1, 2) = T$, $M(1, 3) = T$, so there is no y with $\neg M(1, y)$. Thus the $\exists y$ fails for $x = 1$.

(c)

$$\exists x \forall y M(x, y).$$

Answer: True. **Justification:** For $x = 1$, $M(1, 1) = M(1, 2) = M(1, 3) = T$, so there exists an x (namely 1) for which $M(x, y)$ holds for all y .



PROBLEM 4

Translate each English statement into logical expressions. The domain of discourse is the set of all real numbers.

- (a) The reciprocal of every positive number less than one is greater than one.

$$\forall x \left((0 < x \wedge x < 1) \rightarrow \frac{1}{x} > 1 \right).$$

- (b) There is no smallest number.

$$\neg \exists m \forall y (m \leq y).$$

- (c) Every number other than 0 has a multiplicative inverse.

$$\forall x (x \neq 0 \rightarrow \exists y (x \cdot y = 1)).$$



PROBLEM 5

The sets A , B , and C are defined as follows:

$$A = \{\text{tall, grande, venti}\}, \quad B = \{\text{foam, nofoam}\}, \quad C = \{\text{nonfat, whole}\}.$$

Use the definitions for A , B , and C to answer the questions. Express the elements using n -tuple notation.

- (a) Write an element from the set $A \times B \times C$.
(tall, foam, whole)
- (b) Write an element from the set $B \times A \times C$.
(nofoam, venti, nonfat)
- (c) Write the set $B \times C$ using roster notation.

$$B \times C = \{(\text{foam, nonfat}), (\text{foam, whole}), (\text{nofoam, nonfat}), (\text{nofoam, whole})\}.$$